



LAB 02 ANSWERS REPORT

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COMP3742, TONSLEY
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1. Load the Iris dataset and display the first five rows. What are the feature names and target classes?

- Feature names:
 - Sepal length
 - Sepal width
 - Petal length
 - Petal width
- Target names
 - Iris setosa
 - Iris versicolor
 - Iris virginica

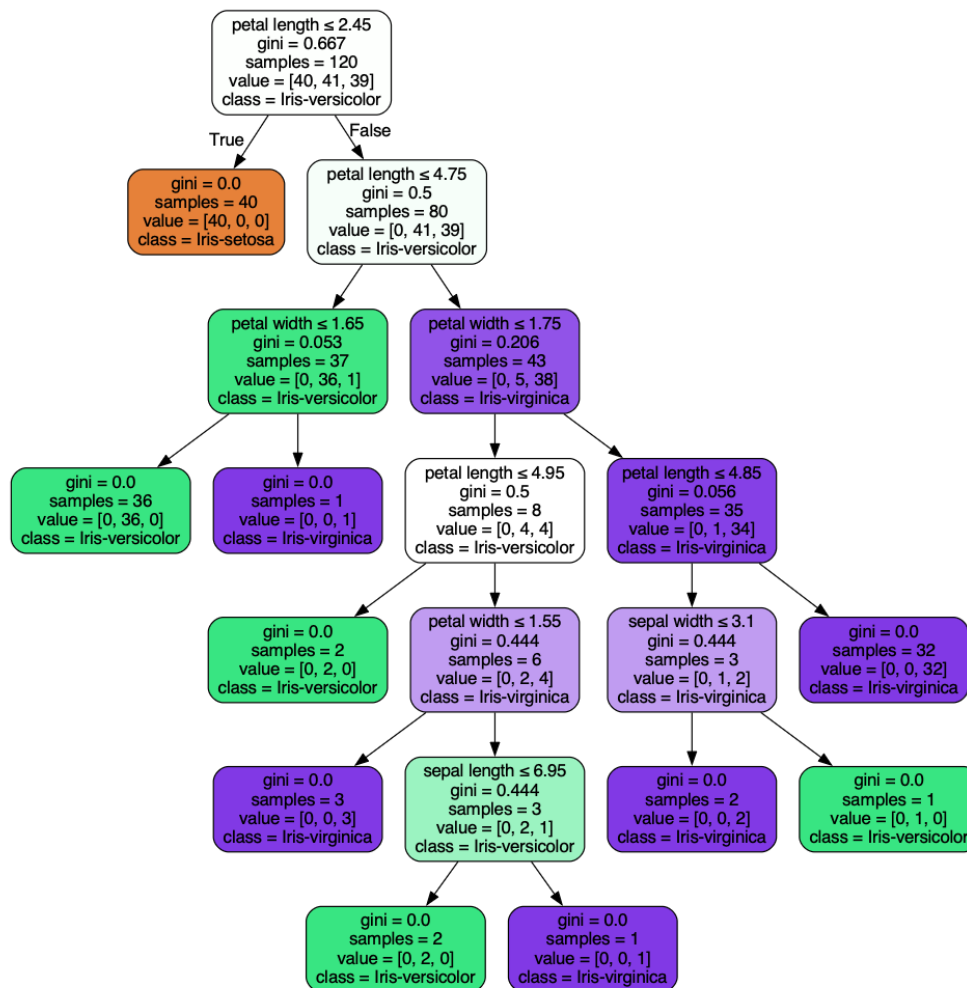
2. Visualize the pairwise relationships between the features of the Iris dataset. What insights can you draw from the scatter plots?

Across several feature combinations, Iris-setosa is more clearly separated from the other classes, which likely suggest it should be relatively easy to classify. In contrast, versicolor and virginica have significant overlap, making them more difficult to distinguish from each other. Petal length, combined with petal width, provided the greatest level of separation between each of the three classes.

3. Train a decision tree classifier on the Iris dataset using scikit-learn. What is the accuracy of this baseline model on the test set?

Baseline module achieves an accuracy of 93%.

4. Visualize the decision tree. What are the most important features according to the decision tree?



Based on the decision tree, the most important feature was petal length, with an importance of 0.9060. In contrast, petal width, sepal length, and sepal width had drastically lower significant scores, ranging from 0.0062 to 0.0586.

5. Perform hyperparameter optimization for this decision tree classifier using GridSearchCV with 5-fold cross-validation. Identify the best hyperparameters and evaluate the model's performance on the test set. Compare the optimized model's performance with the baseline model.

- The best hyperparameters were **'criterion': 'gini', 'max_depth': 30, 'min_samples_leaf': 1, 'min_samples_split': 2}**

- The optimised model achieved an accuracy of 97%, a 4% increase compared to the baseline model.